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\\Research\Investigators\Frederick\ZPL_RETRY\rslh_ep3d_vaso

TA: 14 sec Coil Selection: Auto Voxel Size: 2.5×2.5×2.5 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	48
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.50 mm
TR	130.0 ms
Vol. TR	12.50064 s
TE 1	48.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---

Contrast - Common

TR	130.0 ms
Vol. TR	12.50064 s
TE 1	48.00 ms
Multi-echo spacing	88.00 ms
MTC	Off
Magn. Preparation	Non-sel. IR
TI 1	3130.32 ms
TI 2	9370.32 ms
Flip Angle	13 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Reordering	Linear
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Resolution - Common

FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.50 mm
Base Resolution	84
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	48
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	210 mm
FOV Phase	100.0 %
Slice Thickness	2.50 mm
TR	130.0 ms
Vol. TR	12.50064 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm

Geometry - AutoAlign

Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	210 mm
R >> L	210 mm
F >> H	120 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Sequence - Part 1

Sequence Name	vso rslh6.0
Dimension	3D

Sequence - Part 1

Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1044 Hz/Px
Echo Spacing	1.04 ms
Asymmetric Echo	Off
Turbo Factor	48
Segmentation	1
EPI Factor	84

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	1080 us
RF time x BW	18
PAT ref. FA	5 deg
Fat sat. FA	110 deg
Variable FA	4
Invert PE	Off
Min. TE w/ PF	On
Expert ramping	Off
Trigger per shot	Off
Noise image	Off
Relax spoilers	Off
MT flip angle	170 deg
MT off-res.	2000 Hz
MT RF duration	12800 us
Phase Correction	per Series
EPI ramp factor	1.10
EPI ramp factor 2	1.10
Ramp sampling	1.00 1:max
Slab Scale	0 %
Modify Ice Config	On
G-factor map	Off
Mosaic DICOMs	Off
GRAPPA Regularization	5000 /10^6
G. spoil dephasing[1]	0 pi
G. spoil dephasing[2]	2 pi
G. spoil dephasing[3]	2 pi
Read polarity	Regular RO
VASO Pre-T11 delay	0.00 ms
VASO Pre-T12 delay	0 ms
VASO Pre-Inv delay	0 ms

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators\\Frederick\\ZPL_RETRY\\ZPL_RG_EPSI_FID_v1h

TA: 4:33:55 h Coil Selection: Auto Voxel Size: 1.1×2.2×1.0 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	240 mm
FOV R >> L	240 mm
FOV F >> H	72 mm
Vol A >> P	240 mm
Vol R >> L	240 mm
Vol F >> H	60 mm
TR	1000.0 ms
TE	1.60 ms
Averages	1

Contrast - Common

TR	1000.0 ms
TE	1.60 ms
Ernst angle	61 deg
Prep. pulses	6 #
T1 Estimated	1400 ms
T1 WaterSupp	1000 ms
WaterSupp Angle	90 deg
FOV Pos, Read	0 um
FOV Pos, Phase	0 um
FOV Pos, Slice	0 um
Flip Angle	27 deg
Preparation Scans	10
Averages	1
Water Suppression	
Water Suppr. BW	80 Hz

Resolution - Common

FOV A >> P	240 mm
FOV R >> L	240 mm
FOV F >> H	72 mm
Scan Res. A >> P	216

Resolution - Common

Scan Res. R >> L	110
Scan Res. F >> H	72
Interpol. Res. A >> P	256
Interpol. Res. R >> L	128
Interpol. Res. F >> H	128
Hamming	Off
Dimension	3D
Phase Encoding	Full
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	240 mm
FOV R >> L	240 mm
FOV F >> H	72 mm
Vol A >> P	240 mm
Vol R >> L	240 mm
Vol F >> H	60 mm
Fully Excited Vol	Off
Saturation Region	1
Thickness	30.00 mm
Position	Isocenter
Orientation	Transversal
Sat. Delta Frequ.	0.00 ppm
Saturation Region	2
Thickness	30.00 mm
Position	Isocenter
Orientation	Transversal
Sat. Delta Frequ.	0.00 ppm

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L

System - Miscellaneous

Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	240 mm
R >> L	240 mm
F >> H	60 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.249403 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Sequence - Common

Sequence Name	ZPL_epsilon_fid_v1h
Preparation Scans	10
Delta Frequency	-2.40 ppm
Dimension	3D
Save Uncombined	Off
Bandwidth	167000 Hz
Acquisition Duration	0 ms
Remove Oversampling	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

PE Samp EPSI	No Customized Sampling Pattern
PE Samp EBSI	No Customized Sampling Pattern
PE Samp DEPI	No Customized Sampling Pattern

Sequence - Special

Mode: DEPI nav	0 #
Mode: DEPI ref	0 #
Mode: MotionNav	60 #
Mode: WatSupNav	1 #
Mode: BipolNav	1 #
Mode: RFspoil	0 #
Ramping time (PE)	120 us
Grad duration (PE)	600 us
Blip ramp time (PE)	60 us
Blip ramp time (SL)	60 us
Jump ramp time (JP)	170 us
Jump flat time (JP)	200 us
Ramping time (SP)	180 us
Spoil duration (SP)	4000 us
Spoil amplitude (SP)	15 mT/mm
EPSI. Num: Echo	74 #
EPSI. Num: Shift	0 #
EPSI. Ramp. Samp	16 #
EPSI. RO. Ramp time	110 us
EPSI. EchoSpace	1760 us
EPSC. Num: Read	110 #
EPSC. Num: Phas	24 #
EPSC. Num: Slic	24 #
EPSC. Num: Echo	48 #
EPSC. Ramp. Samp	16 #
EPSC2. Num: Phas	6 #
EPSC2. Num: Slic	6 #
EPSC. EchoSpace	1760 us
Fnav. Read points	100 #
Fnav. Dwell time	10 us
Mnav. Echo Pair	10 #